Code for "Subspace reverse-correlation estimation of receptive fields during free viewing"

To use the code,
1. Make sure the data files are under ./data/ directory.
2. Open eye_cal.m file and specify the experiment name.
3. Run eye_cal following the instructions in the code

The experiment databases with the .db extensions include a Matlab table named "log" with the following structure:

frame	sign	kx	ky	sbxframe	xpos	ypos	area	speed	signal
0	-1	0	-10	4.527e+05	2.3999	-0.13561	191.25	0	{10×505 double}
12	-1	-4	-9	4.5818e+05	2.4495	-0.15445	190.9	0	{10×505 double}
.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.

In this data structure, each row corresponds to a trial in the experiment.

The k,l, and s parameters in the paper specify the stimuli. In the above data set, k="kx", l="ky", s="sign".

"xpos" and "ypos" are the eye positions in the camera coordinates.

"signal" specifies the neuronal responses. In each cell, there are 10 time bins (each trial is 200ms, divided into 20ms intervals) and N neurons.
In the example above, we have 505 neurons.